

Lecture 04: Functions

Understand functions, function calls, built-in function, string formatting

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Motivation example

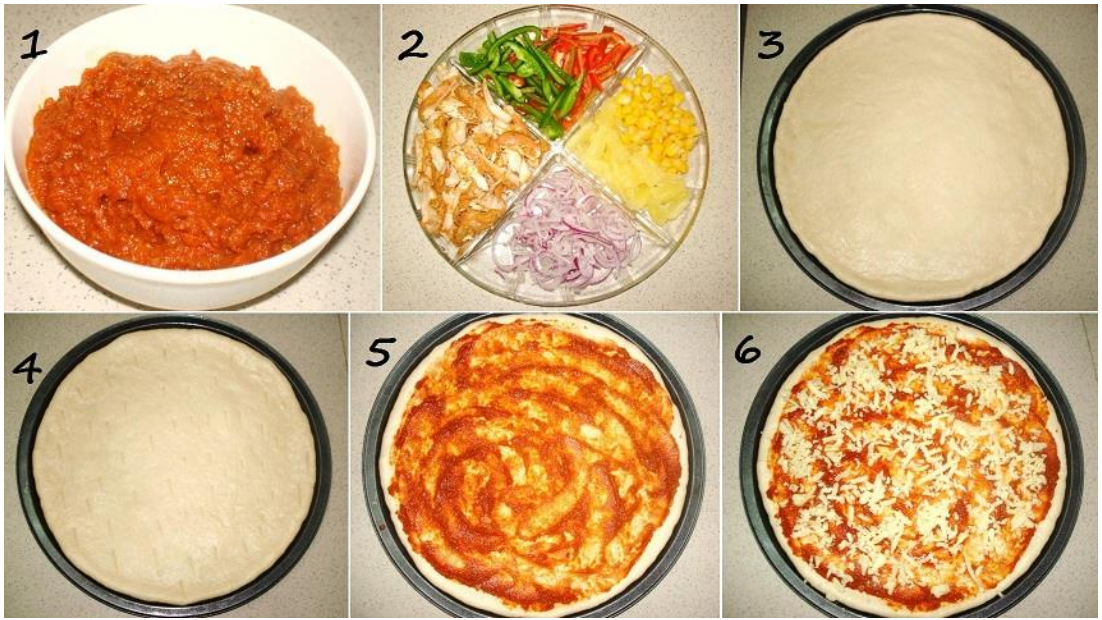
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How to make a Perfect Pizza step-by-step?



Motivation example

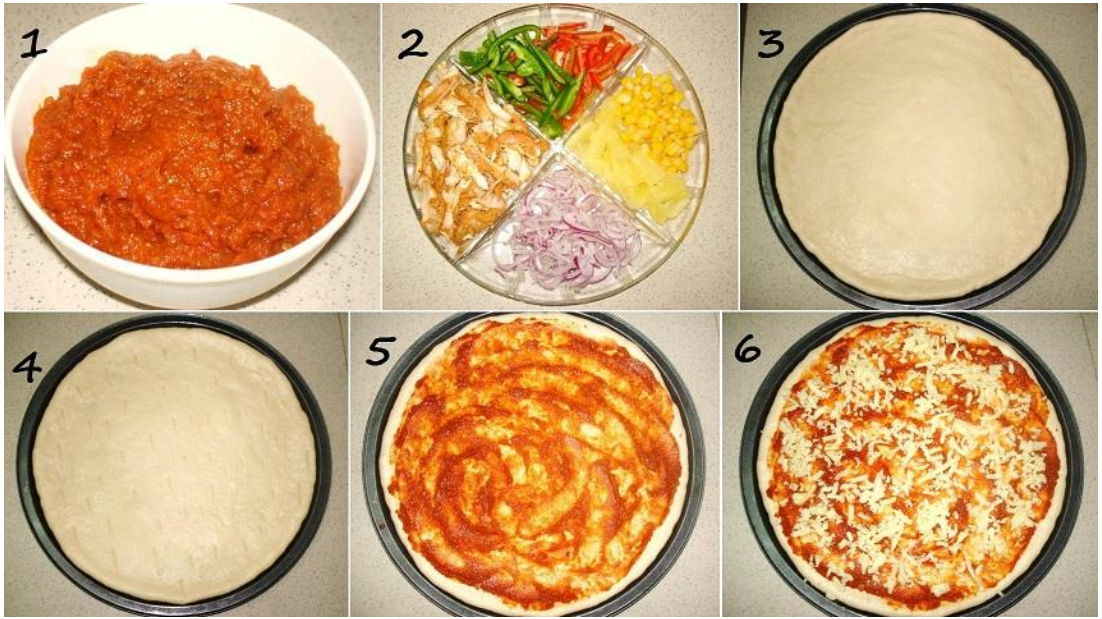
Making a Perfect Pizza step by step:



Step 1: Prepare the pizza sauce.

Motivation example

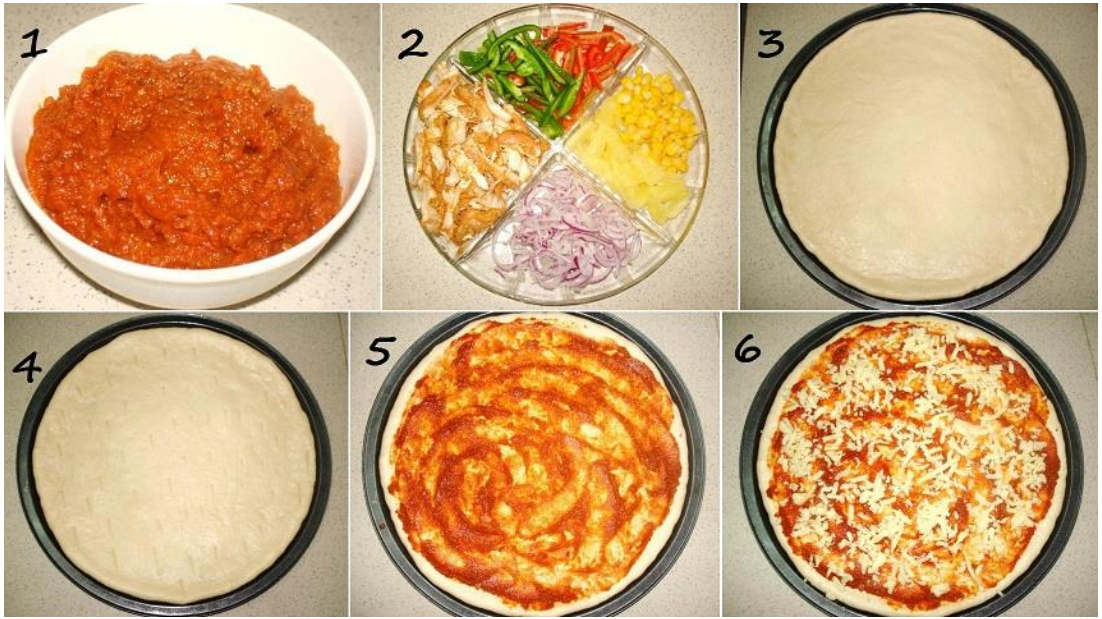
Making a Perfect Pizza step by step:



Step 1: Prepare the pizza sauce.
Step 2: Get all your toppings ready.

Motivation example

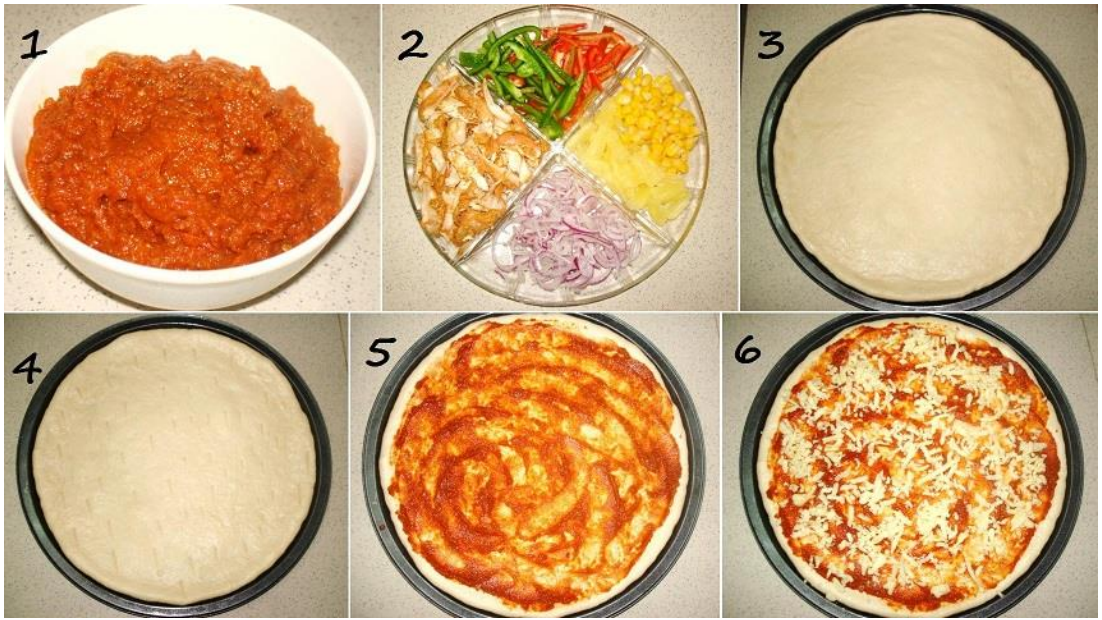
Making a Perfect Pizza step by step:



Step 1: Prepare the pizza sauce.
Step 2: Get all your toppings ready.
Step 3: Now prepare the dough.

Motivation example

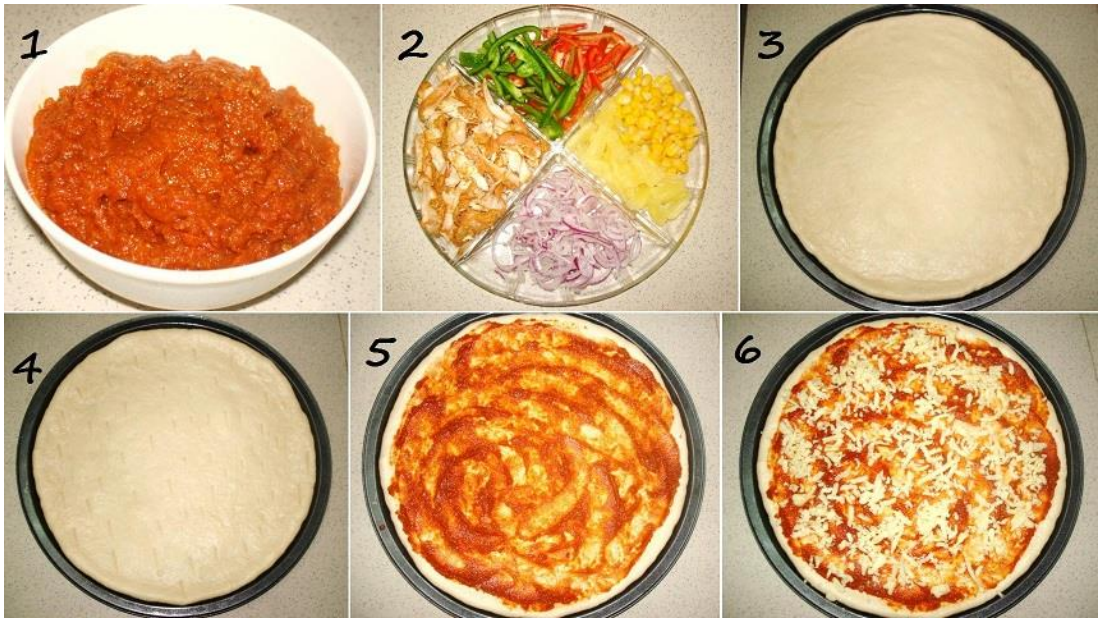
Making a Perfect Pizza step by step:



Step 1: Prepare the pizza sauce.
Step 2: Get all your toppings ready.
Step 3: Now prepare the dough.
Step 4: Take a sharp knife and make tiny slits all over the base.

Motivation example

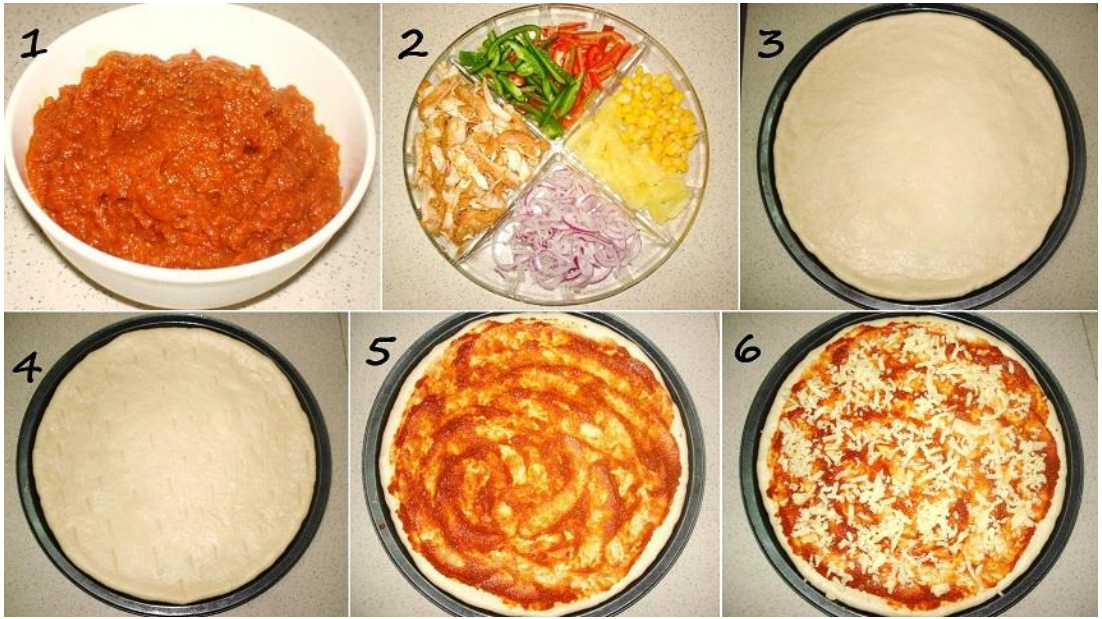
Making a Perfect Pizza step by step:



Step 1: Prepare the pizza sauce.
Step 2: Get all your toppings ready.
Step 3: Now prepare the dough.
Step 4: Take a sharp knife and make tiny slits all over the base.
Step 5: Apply a coating of your pizza sauce as evenly as you can.

Motivation example

Making a Perfect Pizza step by step:



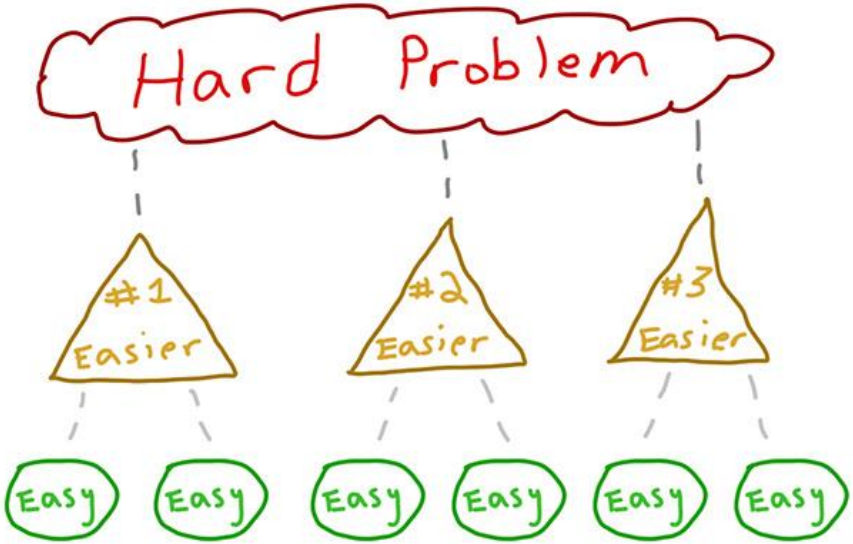
Step 1: Prepare the pizza sauce.
Step 2: Get all your toppings ready.
Step 3: Now prepare the dough.
Step 4: Take a sharp knife and make tiny slits all over the base.
Step 5: Apply a coating of your pizza sauce as evenly as you can.
Step 6: Now add some cheese, sprinkle a light layer all over, and then add your toppings one at a time.

Question

What have you learned regarding the method we used to make a pizza?

Breaking Down Problems

- Dividing a complex problem into smaller chunks makes your problem/program easy to understand and reuse.
- Breaking down problems is the #1 software developer skill.



Source

Outline: Plan for today

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Learning outcomes: Upon the completion of this lecture, students will be able to:

- understand function characteristics and function calls in Python.
- use the standard library in Python.
- write and use built-in functions.
- introduce external libraries in Python.

Functions (mathematics)

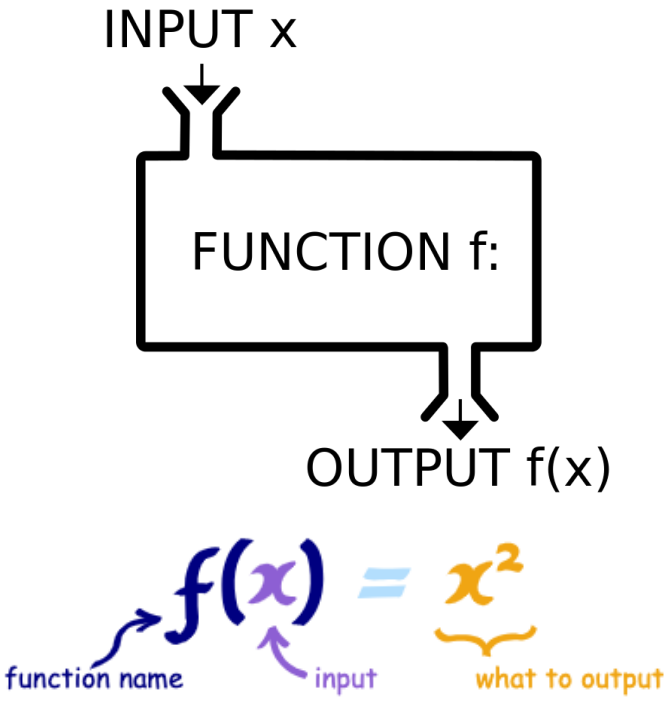
In math, a function is a rule or relationship that connects each input to exactly one output.

- A domain (the set of all possible inputs)
- A codomain (the set of possible outputs)

then a function f is a mapping:

$$f: \text{Domain} \longrightarrow \text{Codomain}$$

such that each element of the domain is assigned to exactly one element in the codomain.



Source

Functions (programming)

- A function is a block of code that performs a specific task.
- Functions are reusable.
- Only run when they are "called" in a program.
- Characteristics:
  - Has a name
  - Has parameters (null or more)
  - Has a docstring (recommended)
  - Has a body
  - Can return something

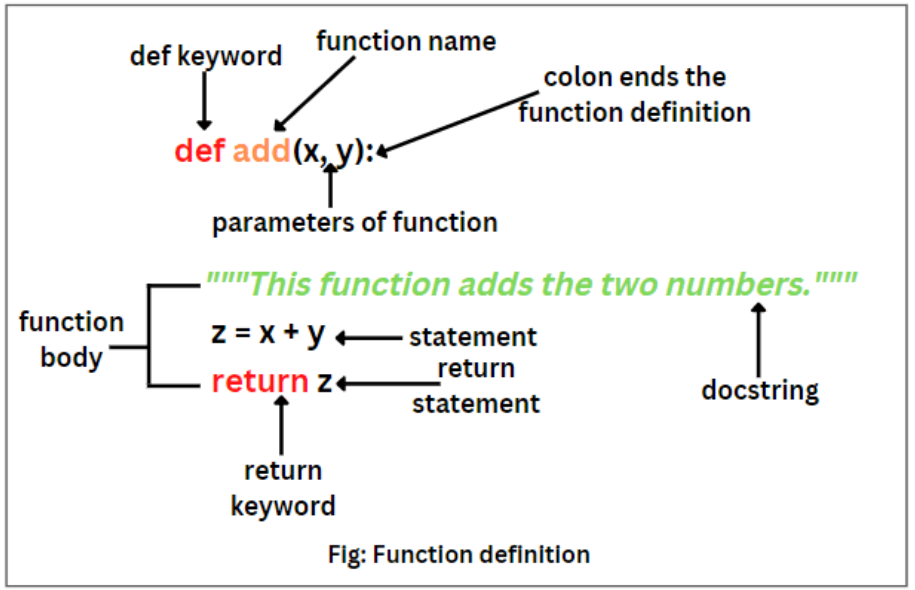


Fig: Function definition

Functions (programming)

- Keyword **def** marks the start of the function header.
- A **function name** to uniquely identify it.
- **Parameters** through which we pass values to a function. They are optional.
- A **colon (:)** to mark the end of the function header.
- Optional documentation string (**docstring**) to describe what the function does.
- An optional **return statement** to return a value from the function.

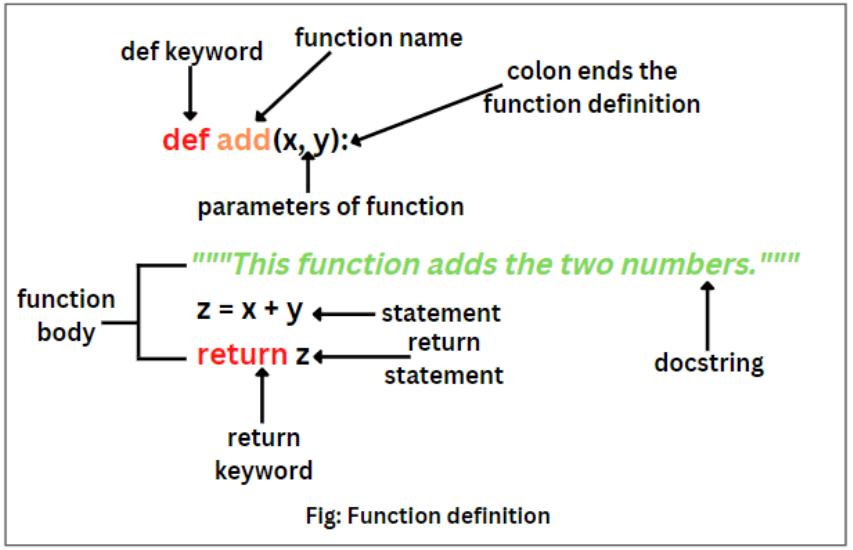


Fig: Function definition

Class practice

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Identify all characteristics of the following function definition

```
def areaTriangle(base, height):
    """
    Input: base and area are positive integer
    Return the area of a triangle using formula area
    area = 0.5 x base x height
    """
    print("The area of your triangle is:")
    return 0.5 * base * height
```

Types of function

There are two types of functions in Python programming:

- **Standard library functions** - These are built-in functions in Python that are available to use.
- **User-defined functions** - We can create our own functions based on our requirements.

The return statement

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- A Python function will always have a return value.
- Format: `return<expression>`
- Any statement after the return is ignored

```
def areaRectangle(length, width):
    area = length * width
    print("The area of your rectangle is:")
    return area
```

Important: If you don't explicitly use a return value in a return statement, or if you totally omit the return statement, then Python will implicitly return a default value for you. That default return value will always be None.

```
def areaRectangle(length, width):
    area = length * width
    return
```

OR

```
def areaRectangle(length, width):
    area = length * width
```

The return statement

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Example:

Python >>>

```
>>> def add_one(x):
...     # No return statement at all
...     result = x + 1
...

>>> value = add_one(5)
>>> value

>>> print(value)
None
```

Let's see an example


```
def greet():
    print('Hello World!')
```

Here:

- We have created a function named `greet()`. It simply prints the text Hello World!
- This function doesn't return any values.

Calling a Function in Python

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Declare a function named "`greet()`"

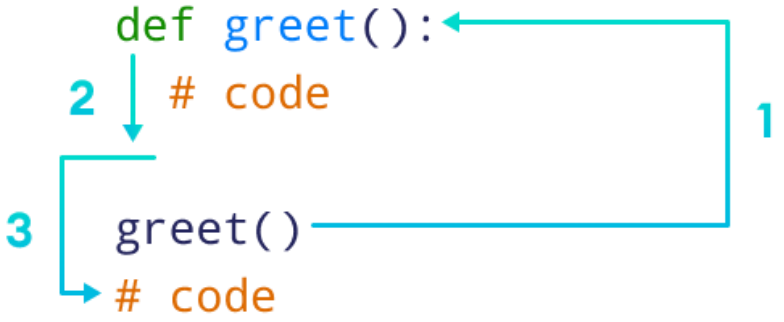
```
def greet():
    print('Hello World!')
```

Now, to use this function, we need to call it. Here's how we can call the `greet()` function in Python.

```
# call the function
greet()
```

How does it work?

- When the function is called, the control of the program goes to the function definition.
- All codes inside the function are executed.
- The control of the program jumps to the next statement after the function call.



Class practice

Write a function to calculate the area of a circle, given by $A = \pi r^2$

Class practice

Write a function to calculate the area of a circle, given by $A = \pi r^2$

Sample code:

```
1 def area_circle(radius):
2     pi = 3.14159
3     area = pi * radius**2
4     return area
5
6 print(area_circle(5))
```

Output: 78.53975

Calling a built-in function in Python

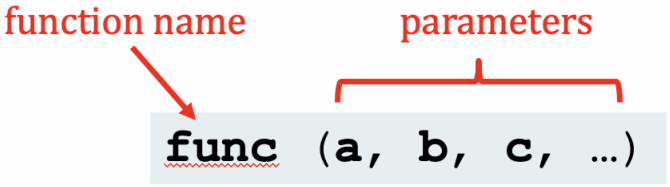
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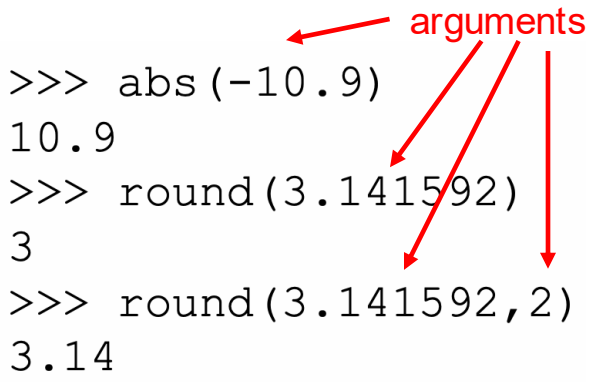
A collection of script modules accessible to a Python program. It can be used by 'calling/importing' them at the beginning of a script.

		Built-in Functions		
abs()	delattr()	hash()	memoryview()	set()
all()	dict()	help()	min()	setattr()
any()	dir()	hex()	next()	slice()
ascii()	divmod()	id()	object()	sorted()
bin()	enumerate()	input()	oct()	staticmethod()
bool()	eval()	int()	open()	str()
breakpoint()	exec()	isinstance()	ord()	sum()
bytearray()	filter()	issubclass()	pow()	super()
bytes()	float()	iter()	print()	tuple()
callable()	format()	len()	property()	type()
chr()	frozenset()	list()	range()	vars()
classmethod()	getattr()	locals()	repr()	zip()
compile()	globals()	map()	reversed()	__import__()
complex()	hasattr()	max()	round()	

Calling a built-in function in Python

Function form:





Some of the built-in functions you have used require arguments: the values that control how the function does its job.

```
>>> x = math.log10(100)
```

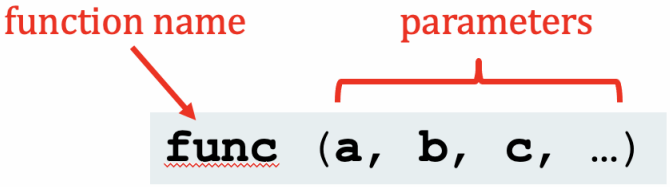
log10 takes numeric value as an argument

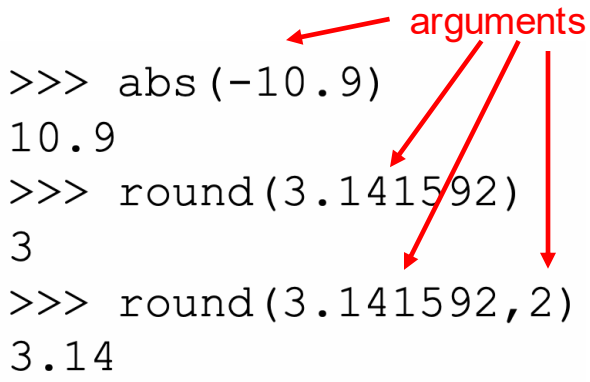
```
>>> y = math.pow(2, 3)
```

pow takes two arguments, the base and the exponent

Calling a built-in function in Python

Function form:





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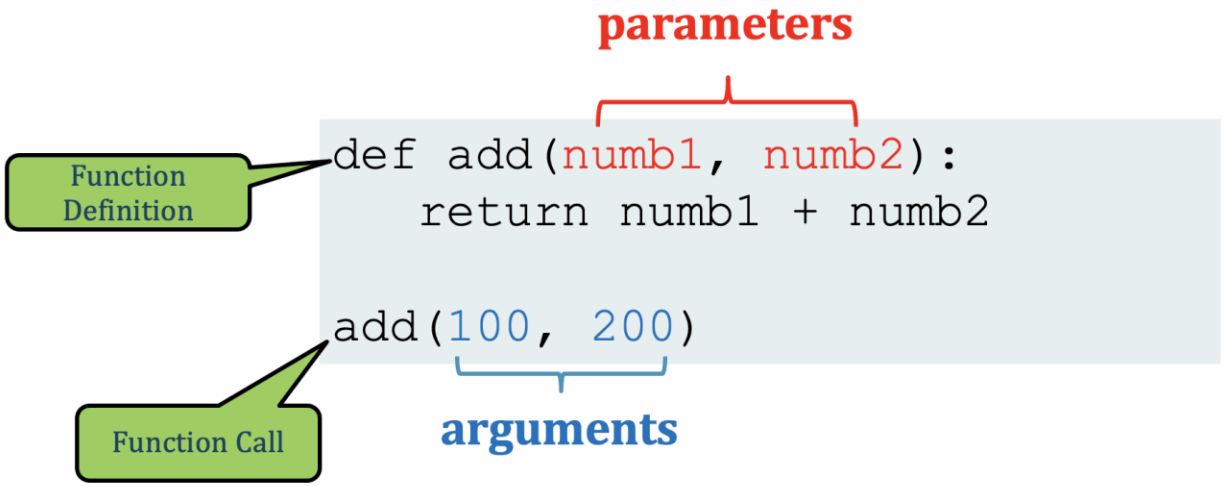
```
>>> y = math.pow(2, 3)
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pow takes two arguments, the base and the exponent

Q: What are the differences between parameters and arguments?

Parameters vs. Arguments

A parameter is the variable listed inside the parentheses in the function definition. An argument is the value that are sent to the function when it is called.



Formatting strings

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- Objective: To format a targeted value and insert it inside the string placeholder `{}`.
- Syntax: `string.format(value)`

```
>>> msg1 = "Welcome to {place}!".format(place = "VinUni")
>>> print(msg1)
Welcome to VinUni!

>>> msg2 = "Hi {0}, are you okay?".format("Billy")
>>> print(msg2)
Hi Billy, are you okay?

>>> msg3 = "Hello {}, your exam score is {}.".format("Ali
Baba", 88)
>>> print(msg3)
Hello Ali Baba, your exam score is 88.
```

Formatting strings

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```
>>> message = "Hi {}, you made me {}!"
>>> subject = "Python"
>>> feeling = "crazy"
>>> print(message.format(subject,feeling))
Hi Python, you made me crazy!

>>> print("{1}, {2}, and {3} are humans?".format("Billy", "Thomas",
"Kok-Seng"))
IndexError: Replacement index 3 out of range for positional args
tuple

>>> print("{2}, {0}, and {1} are humans?".format("Billy", "Thomas",
"Kok-Seng"))
Kok-Seng, Billy, and Thomas are humans?
```

Formatting strings

```
>>> sale1 = "Total is {:,}vnd.".format(500000)
>>> print(sale1)
Total is 500,000vnd.
```

The :, adds commas every three digits.

```
>>> sale2 = "Total is {:f}vnd.".format(500000)
>>> print(sale2)
Total is 500000.000000vnd.
```

The :f formats the number as a float, with 6 decimal places by default.

```
>>> sale3 = "Total is {:.2f}vnd.".format(500000)
>>> print(sale3)
Total is 500000.00vnd.
```

The :.2f means 2 digits after the decimal point.

External Libraries

- **Numpy** is a general-purpose array-processing package. It provides a high-performance multidimensional array object.
- **Pandas** is the most popular Python library that is used for data analysis.
- **Matplotlib** is a plotting library for the Python programming language and its numerical mathematics extension NumPy.



External Libraries - numpy

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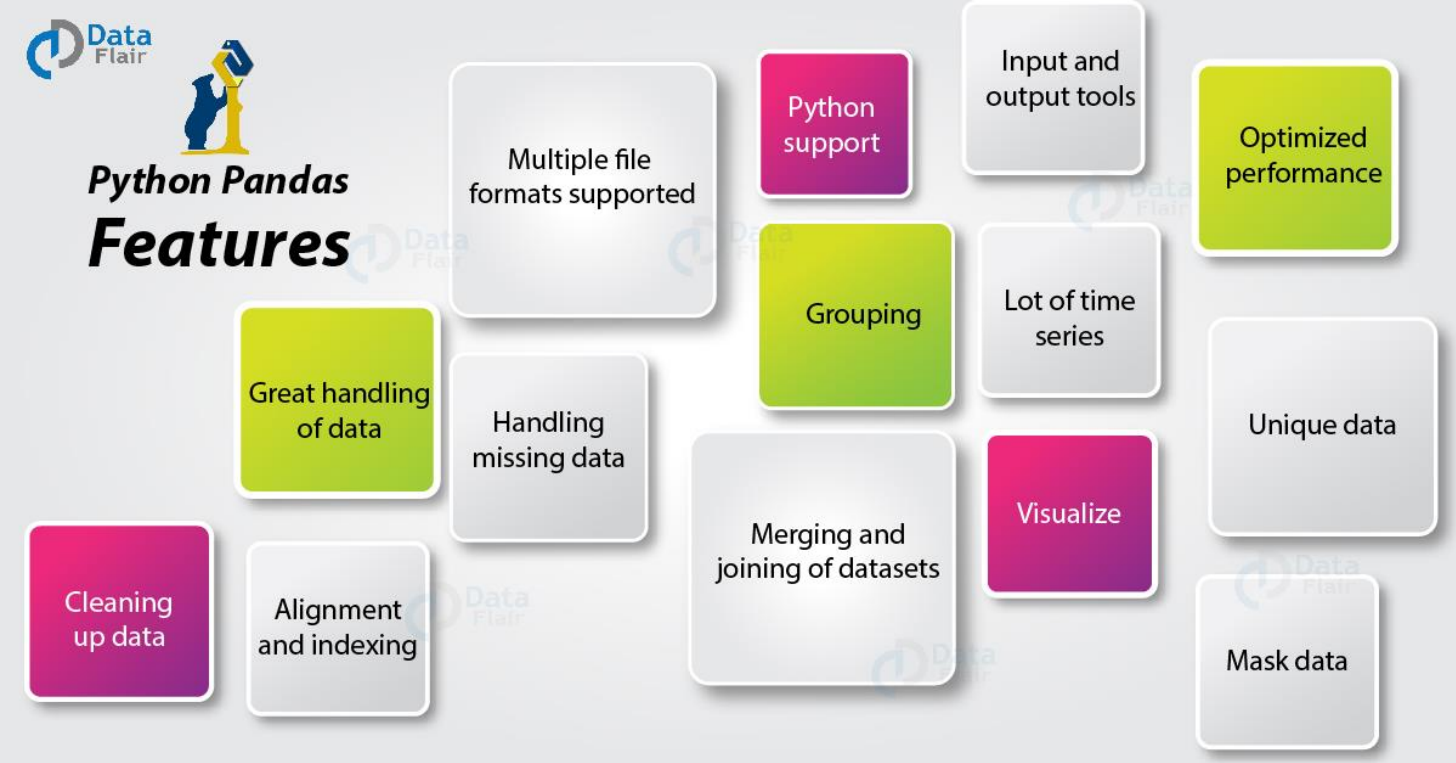
NumPy brings the computational power of languages like C and Fortran to Python, a language much easier to learn and use. With this power comes simplicity: a solution in NumPy is often clear and elegant.

Quantum Computing	Statistical Computing	Signal Processing	Image Processing	3-D Visualization	Symbolic Computing	Astronomy Processes	Cognitive Psychology
							
QuTiP PyQuil Qiskit	Pandas statsmodels Seaborn	SciPy PyWavelets	Scikit-image OpenCV	Mayavi Napari	SymPy	AstroPy SunPy SpacePy	PsychoPy

Bioinformatics	Bayesian Inference	Mathematical Analysis	Simulation Modeling	Multi-variate Analysis	Geographic Processing	Interactive Computing
						
BioPython Scikit-Bio PyEnsembl	PyStan PyMC3	SciPy SymPy cvxpy FEniCS	PyDSTool	PyChem	Shapely GeoPandas Folium	Jupyter IPython Binder

https://numpy.org/

External Libraries - pandas

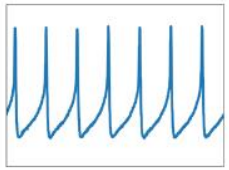
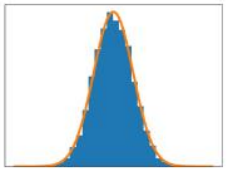
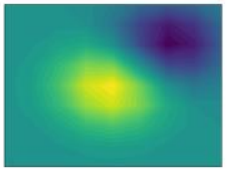
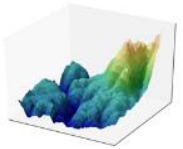


https://pandas.pydata.org/

External Libraries - matplotlib

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Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.

Matplotlib makes easy things easy and hard things possible.

Create

- Develop publication quality plots with just a few lines of code
- Use interactive figures that can zoom, pan, update...

Customize

- Take full control of line styles, font properties, axes properties...
- Export and embed to a number of file formats and interactive environments

Extend

- Explore tailored functionality provided by third party packages
- Learn more about Matplotlib through the many external learning resources

https://matplotlib.org